Grade 8
Unit 1
Lesson 6 Practice

Name_Answer Key_________
Date____________________
Period___________________

1. Solve the equation by applying the properties of equality.

$$-9.6 = \frac{y}{4} - 3\frac{1}{2}$$
$$+3\frac{1}{2} \qquad + 3\frac{1}{2}$$
$$-9.6 + 3.5 = \frac{y}{4}$$
$$-6.1 = \frac{y}{4}$$
$$4 \cdot (-6.1) = \frac{y}{4} \cdot 4$$
$$\boxed{-24.4 = y}$$

2. Use substitution into the original equation using your solution to (1) to verify the solution.

$$-9.6 = \frac{y}{4} - 3\frac{1}{2}$$
$$-9.6 = \frac{-24.4}{4} - 3\frac{1}{2}$$
$$-9.6 = -6.1 - 3.5$$
$$-9.6 = -6.1 + (-3.5)$$
$$-9.6 = -9.6 \checkmark$$

3. Solve the equation by applying the properties of equality.

$$-3c - 6.24 = 18$$
$$+6.24 \quad + 6.24$$
$$\frac{-3c}{-3} = \frac{24.24}{-3}$$
$$\boxed{c = -8.08}$$

4. Use substitution into the original equation using your solution to (3) to verify the solution.

$$-3c - 6.24 = 18$$
$$-3(-8.08) - 6.24 = 18$$
$$24.24 - 6.24 = 18$$
$$18 = 18 \checkmark$$

5. Solve the equation by applying the distributive property first.

$$-14 = \frac{1}{2}(7 - 5m)$$

$-14 = 3.5 - 2.5m$
$-3.5 \quad - 3.5$

$-17.5 = 2.5m$

$\frac{-17.5}{-2.5} = \frac{-2.5m}{-2.5}$

$\boxed{m = 7}$

6. Solve the equation by dividing both sides by the factor being distributed.
Note: You should get the same solution as (5).

$$-14 = \frac{1}{2}(7 - 5m)$$

$\div \frac{1}{2}(-14) = (\frac{1}{2}(7 - 5m)) \div \frac{1}{2}$

$-14 \cdot \frac{2}{1} = (\frac{1}{2}(7 - 5m)) \cdot \frac{2}{1}$

$-28 = 7 - 5m$
$-7 \quad - 7$

$\frac{-35}{-5} = \frac{-5m}{-5}$

$\boxed{7 = m}$

For questions 7-10, use the following scenario.

The water temperature at the beach started at 81.5°F at 4 PM. As it gets later in the day, the temperature drops 0.6 degrees each hour. If the water temperature is now 77.9°F, what time was the temperature measured again?

7. Define the unknown quantity using a variable.

$h =$ number of hours the temperature dropped

(or any variable)

8. Write an equation that can be used to find the unknown quantity using the variable you defined.

$81.5 - 0.6h = 77.9$

9. Solve the equation.

$81.5 - 0.6h = 77.9$
$-81.5 \qquad - 81.5$

$\frac{-0.6\text{h}}{-0.6} = \frac{-3.6}{-0.6}$

$h = 6$

10. Explain what the solution means in the context of the situation.

The solution is the number of hours after 4pm that the new temperature was measured.
To find the new time add 6 and 4 to get 10pm.